Institutional Robustness Under Capability Escalation in Sequential Commons

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Abstract

Advanced AI systems are increasingly expected to operate in plural settings rather than as a single monolithic agent. That shift changes the alignment problem from supervising one system to designing institutions that remain robust when many interacting systems face escalating strategic pressure. We study that question in sequential commons. Fishery Commons is retained as a precursor signal-design study showing that central interventions can be ranked under repeated adversarial strategy injection, with adaptive quotas emerging as the strongest overall top-down signal in the medium tier. Harvest Commons becomes the main architecture study. A restored four-condition pilot shows that bottom-up-only governance is distinguishable from both no governance and top-down-only control, but remains ecologically weaker than architectures with central intervention. A high-power confirmatory matrix then shows that hybrid governance ranks first in all eight decision cells under a stronger search-over-mutations attacker. Relative to top-down-only control, hybrid improves patch health in all eight cells, is non-worse on neighborhood overharvest in six of eight, and incurs a moderate mean welfare cost. A capability-ladder analysis over random, mutation, adversarial-heuristic, and search-based attackers shows that the hybrid-versus-top-down comparison is the more sensitive part of the ordering, whereas both governed architectures retain a stable ecological advantage over local-only control. Welfare-incidence logging further shows that the incremental cost of moving from top-down-only to hybrid governance is modest relative to the much larger penalties associated with local-only governance. These results support a benchmark-level claim: in stylized sequential commons, governance architecture can be evaluated as institutional robustness under capability escalation, and multi-scale governance can outperform central-only control without eliminating trade-offs.

1 Introduction

Many alignment and oversight proposals begin from a dyadic picture: a human or institution supervises one powerful AI system. That framing remains useful for a range of problems, but it becomes incomplete once advanced AI deployment is expected to involve many interacting systems with partly aligned and partly conflicting objectives. Recent arguments about agentic AI emphasize this plural structure directly: intelligence may scale through interacting constellations of agents, services, and institutions rather than through a single monolithic system [2]. On that view, alignment becomes partly an institutional problem. The question is not only whether one agent is well behaved, but whether the surrounding rules, protocols, and governance arrangements remain robust as strategic pressure increases.

This paper develops that framing through sequential commons. Commons settings provide a useful bridge between institutional theory and multi-agent learning because they make the central tension legible: individually beneficial behavior can still damage a shared system unless the rules of interaction are well designed [3, 7, 8]. Work on sequential social dilemmas and multi-agent evaluation already points in this direction by showing that cooperation depends on temporally extended interaction, partner composition, and held-out social situations rather than only on one-shot payoffs [4–6]. Cooperative-AI work strengthens the same point from a game-theoretic angle: individually aligned systems do not guarantee collectively good outcomes, so the interaction structure matters in its own right [1].

The contribution here is therefore framed as *institutional robustness under capability escalation*. In the benchmark, capability escalation is operationalized as repeated adversarial strategy injection: weaker strategies are replaced over generations, and stronger injectors search for more exploitative children. Governance conditions are then interpreted as candidate institutional designs. Under this translation, top-down controls,

bottom-up coordination, and hybrid arrangements are not merely environment toggles. They are alternative ways of organizing restraint, intervention, and response inside a plural system.

The empirical program remains grounded in two linked studies, but their roles are asymmetric. Fishery Commons is the controlled precursor study. It asks which central governance signals survive repeated strategic turnover in a minimal renewable commons. Harvest Commons is the main study. It asks which governance architecture remains robust once local interaction, communication, and local side-payments are allowed to matter. The follow-up therefore reinterprets an existing benchmark around a narrower question: when strategic pressure rises, what do central-only, local-only, and multi-scale governance each contribute?

2 Methods

Both studies use the same core invasion protocol. A population of strategies is evaluated over generations, weaker strategies are removed, and new strategies are injected by a specified generator. Selection pressure is applied in train conditions and robustness is reported in held-out test conditions. This separates adaptation from evaluation and turns governance comparisons into tests of out-of-distribution stability rather than only in-sample fit.

2.1 Study 1: Fishery Commons as a signal-design precursor

Fishery Commons is a minimal renewable commons with one shared stock, per-agent harvest actions, regeneration, and collapse dynamics. Governance changes extraction pressure through monitoring, sanctions, quota clipping, adaptive quotas, or temporary closures. The first Fishery layer is a matched baseline that compares no governance against monitoring with sanctions under two injector families: mutation and live JSON-based LLM injection. The second layer is a broader non-LLM top-down comparison in the medium tier that evaluates monitoring with sanctions, adaptive quotas, and temporary closures across balanced and adversarial-heavy partner mixes, adversarial pressures 0.3 and 0.5, and three injector families: mutation, an adversarial heuristic injector, and search over mutations. A medium-tier PPO self-play validation is retained as a policy-family robustness check on the resulting ranking.

2.2 Study 2: Harvest Commons as the main architecture study

Harvest Commons extends the commons problem from one global stock to local renewable patches with local interaction, optional communication, and optional side-payments. Agents act through parameterized local harvest, restraint, request, offer, and compliance components, while governance is imposed by the environment. Across the broader project history, the architecture space includes `none`, `bottom_up_only`, `top_down_only`, and `hybrid`. In this follow-up, those conditions are interpreted as institutional designs rather than as simple environment toggles.

The architecture evidence is organized in two stages. The first is an architecture-restoration pilot over `none`, `bottom_up_only`, `top_down_only`, and `hybrid` in the medium and hard tiers, balanced and adversarial-heavy partner mixes, pressure 0.3, and two non-LLM attackers: mutation and search over mutations. The second is the high-power confirmatory comparison over `bottom_up_only`, `top_down_only`, and `hybrid` under the stronger `search_mutation` attacker in the medium and hard tiers, balanced and adversarial-heavy partner mixes, and pressures 0.3 and 0.5.

The confirmatory ranking rule is lexicographic: lower garden failure first, then higher patch health, then higher welfare. Accordingly, a condition "wins" a decision cell only in the sense of first rank under that rule. Pairwise contrasts are reported for `hybrid - top_down_only`, `hybrid - bottom_up_only`, and `top_down_only - bottom_up_only`.

2.3 Welfare incidence and capability ladder

Harvest evaluation logs per-agent realized welfare, requested harvest, realized harvest, prevented harvest, credit sent, credit received, local patch health, and whether the agent was targeted or capped. These logs support two incidence views. The first groups agents into low-, middle-, and high-aggression terciles using aggressive-request behavior, with requested harvest as a fallback grouping variable. The second groups

Table 1: Fishery Commons precursor summary. Collapse reductions are measured relative to no governance in the medium-tier non-LLM sweep. The final column reports the welfare advantage of adaptive quotas over temporary closures.

Injector	Sanctions	Adaptive quota	Closure	Adaptive quota minus closure welfare
Mutation	0.3737	0.9167	0.9167	2408.81
Adversarial heuristic	0.0208	0.9167	0.9167	2211.61
Search over mutations	0.2617	0.9167	0.9167	2723.02

agents by whether they were targeted for caps during the episode. These summaries are used to distinguish concentrated restraint from broad welfare taxation.

Attacker strength is represented by a non-LLM capability ladder that progresses from random injection to mutation, then to an adversarial heuristic attacker, and finally to search over mutations. The ladder pilot evaluates `bottom_up_only`, `top_down_only`, and `hybrid` in the medium and hard tiers, balanced and adversarial-heavy partner mixes, pressures 0.3 and 0.5, and three runs per cell. In the follow-up, these attackers are interpreted as increasing rungs of adversarial capability. The ladder analysis records the first rung at which a pairwise architecture contrast loses its patch-health advantage, the first rung at which it loses its neighborhood-overharvest advantage, and the first rung at which the welfare cost crosses the -0.15 threshold.

3 Results

3.1 Fishery Commons as a signal-design precursor

The Fishery Commons study is retained as the controlled precursor rather than the center of gravity of the follow-up. Its role is to show that the benchmark can rank central interventions under repeated strategic turnover. In the matched baseline, monitoring with sanctions reduces held-out collapse relative to no governance under both mutation and live JSON-based LLM injection. In the broader non-LLM top-down sweep, adaptive quotas emerge as the strongest overall signal in the medium tier. Across balanced and adversarial-heavy populations, pressures 0.3 and 0.5, and three non-LLM injector families, adaptive quotas win every injector block while temporary closures remain a credible ecological fallback. A medium-tier PPO validation recovers the same qualitative ordering. The precursor conclusion is therefore narrow but informative: central governance signals can be ranked under capability escalation, and adaptive quotas are the strongest top-down intervention in the simpler substrate.

3.2 Architecture restoration

The architecture-restoration pilot broadens Harvest Commons from the earlier narrow pairwise comparison to the full architecture spectrum. In the pilot, `hybrid` ranks first in 6 of 8 decision cells and `top_down_only` ranks first in the remaining 2. Neither `bottom_up_only` nor `none` ranks first. This already answers the first pilot question: `bottom_up_only` is not redundant with `none`. Relative to `none`, bottom-up-only governance improves patch health in 6 of 8 cells and is non-worse on neighborhood overharvest in 6 of 8, although its mean welfare delta is negative. Local coordination therefore contributes something real, but not enough to recover the ecological control achieved by architectures with central intervention.

The broader pilot also clarifies what hybrid governance adds. Relative to `top_down_only`, the pilot yields positive patch-health deltas in 6 of 8 cells and non-worse neighborhood-overharvest deltas in 6 of 8, with a mean welfare delta of only -0.0572. Relative to `bottom_up_only`, both governed architectures buy large patch-health gains but at much larger welfare cost. The pilot interpretation is therefore that local-only governance is empirically distinct, top-down control is ecologically necessary, and the central question is whether hybrid governance adds enough to justify itself over central-only control.

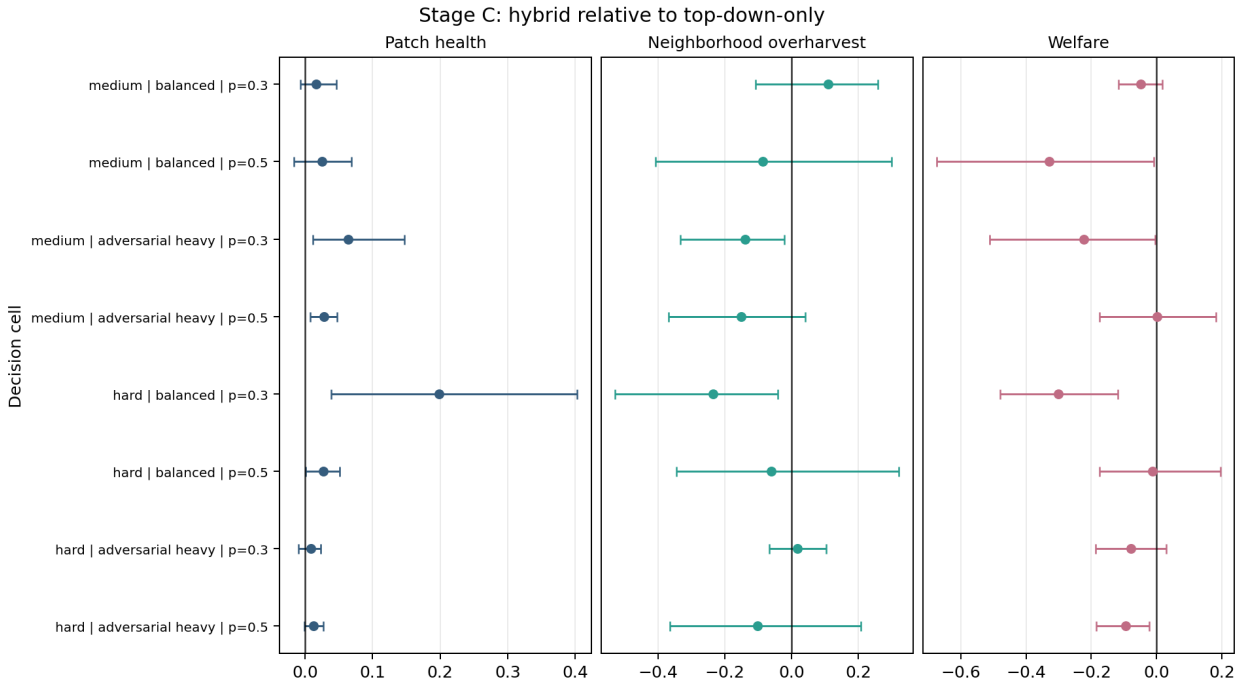

Figure 1: Cell-level Stage C effect sizes for `hybrid - top_down_only` under `search_mutation`. The high-power matrix covers the medium and hard tiers, balanced and adversarial-heavy populations, and pressures 0.3 and 0.5. Hybrid governance ranks first in every decision cell under the failure-then-patch-health-then-welfare rule, with the margin carried by patch health, neighborhood overharvest, and a moderate welfare cost.

3.3 High-power architecture confirmation

The confirmatory Stage C matrix answers that question cleanly. Under the lexicographic failure-then-patch-health-then-welfare rule, `hybrid` ranks first in all 8 decision cells. `top_down_only` ranks second in every cell, and `bottom_up_only` ranks third in every cell.

The most important comparison is `hybrid - top_down_only`. In the high-power matrix, hybrid improves patch health in all 8 cells, is non-worse on neighborhood overharvest in 6 of 8 cells, and has a mean welfare delta of -0.1363. That welfare cost remains inside the planned -0.15 threshold, so the confirmatory gate is satisfied. The mean patch-health delta is $+0.0477$, and the mean neighborhood-overharvest delta is -0.0813. The architecture result is therefore not that hybrid dominates every metric in every cell. It is that, under stronger search-based capability escalation, hybrid is the best overall architecture by the study's ranking rule and achieves that rank primarily through ecological and behavioral margins rather than through a collapse-vs-no-collapse effect.

The full architecture spectrum clarifies the mechanism underlying that result. Relative to `bottom_up_only`, both `hybrid` and `top_down_only` improve patch health in all 8 cells, but both do so with large welfare costs. In the confirmatory matrix, `hybrid - bottom_up_only` has mean patch-health delta $+3.1234$ and mean welfare delta -1.2231, while `top_down_only - bottom_up_only` has mean patch-health delta $+3.0757$ and mean welfare delta -1.0868. Local-only governance therefore retains more welfare on average, but it does so by tolerating weaker ecological control and nonzero failure risk. Central intervention is necessary, and hybrid governance then improves on central-only control enough to rank first in every confirmatory cell.

3.4 Capability ladder

The capability ladder provides an additional robustness test on top of the Stage B and Stage C architecture results. Across the 32 ladder cells, `hybrid` ranks first in 25 and `top_down_only` ranks first in the remaining 7. By injector rung, hybrid wins all 8 random cells, 6 of 8 mutation cells, 4 of 8 adversarial-heuristic cells, and 7 of 8 search-over-mutations cells. The ladder therefore does not support a claim that one architecture dominates unconditionally at every rung. It does show that the architecture ordering remains informative

Table 2: Harvest Commons architecture summary. Stage B is the restoration pilot over four governance conditions; Stage C is the high-power confirmatory comparison over `bottom_up_only`, `top_down_only`, and `hybrid`. Positive patch-health deltas favor the left condition; nonpositive overharvest deltas are improvements for the left condition.

Stage	Contrast	Cells	Patch > 0	Overharvest ≤ 0	Mean welfare
B	bottom_up_only − none	8	6	6	-0.3425
B	hybrid − top_down_only	8	6	6	-0.0572
C	hybrid − top_down_only	8	8	6	-0.1363
C	hybrid − bottom_up_only	8	8	0	-1.2231
C	top_down_only − bottom_up_only	8	8	0	-1.0868

under rising pressure, and that the vulnerable comparison is specifically `hybrid` versus `top_down_only`, not governed architectures versus local-only governance.

That asymmetry is explicit in the first-break table. For `hybrid - bottom_up_only`, the ecological break never occurs in any of the 8 decision cells: the first ecological-break rung is `none` in all 8. The same is true for `top_down_only - bottom_up_only`. In other words, both governed architectures retain their patch-health advantage over local-only governance throughout the full non-LLM attacker ladder. The break points appear elsewhere. In the `hybrid - top_down_only` comparison, the ecological margin survives through the full ladder in 3 of 8 cells, first breaks at mutation in 2, at the adversarial-heuristic rung in 2, and at search over mutations in 1. The control margin follows a similar pattern: no control break in 3 cells, mutation as the first break in 1, adversarial heuristic in 3, and search over mutations in 1.

The ladder also clarifies the role of welfare. The costly-robustness flag trips earlier than the ecological break in the `hybrid - top_down_only` contrast: the first welfare-threshold crossing appears already at the random rung in 6 of 8 cells, at search over mutations in 1, and never in 1. By contrast, the local-only comparisons are ecologically decisive but welfare-expensive from the start. For `hybrid - bottom_up_only`, the first costly-robustness rung is random in all 8 cells, and for `top_down_only - bottom_up_only` it is random in 2 and mutation in 6. The ladder interpretation is therefore not that hybrid governance is globally strongest in every sense. It is that both governed architectures consistently dominate local-only governance ecologically, while the finer hybrid-versus-top-down margin remains real but more sensitive to attacker family and welfare cost.

3.5 Welfare incidence

The incidence summaries help interpret the welfare trade-off. For the confirmatory `hybrid - top_down_only` comparison, the per-tercile welfare deltas are modest across the board: averaged over the high-, mid-, and low-aggression groups, the mean welfare effects are approximately -0.0134, -0.0168, and -0.0379, respectively. The targeted-agent view tells a similar story. For agents who are actually targeted by caps, the mean welfare delta is only -0.0227, while the corresponding mean local-patch-health delta is $+0.0468$. In other words, the incremental cost of moving from central-only to hybrid governance is not large in the same way as the architecture-versus-local-only contrast.

This distinction is important for interpretation. The larger welfare penalty in this study is the cost of restoring ecological control when moving away from local-only governance, not the cost of adding local coordination on top of an already functioning central architecture. Hybrid governance therefore appears less as a globally punitive architecture than as a modestly more interventionist extension of top-down control that yields additional ecological stability.

4 Discussion

This follow-up refines the contribution of the existing benchmark rather than broadening it indiscriminately. The earlier two-study paper already showed that governance can be evaluated under repeated adversarial strategy injection in sequential commons, that central signals can be ranked in a minimal setting, and that a narrower hybrid-versus-top-down comparison remains informative under a stronger attacker. The Harvest-led follow-up makes the architecture question explicit. Once the full architecture spectrum is restored,

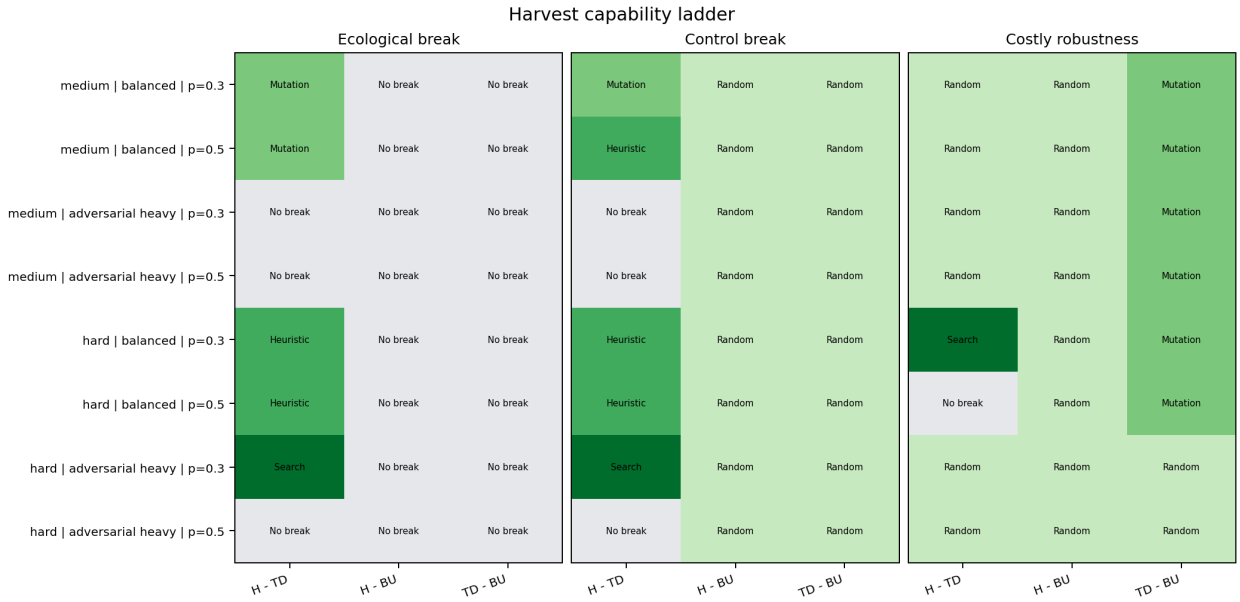

Figure 2: Capability-ladder summary for Harvest Commons. Each cell reports the first attacker rung at which a pairwise architecture contrast loses its ecological advantage, loses its neighborhood-overharvest advantage, or crosses the welfare-cost threshold. Abbreviations are `H` for `hybrid`, `TD` for `top_down_only`, and `BU` for `bottom_up_only`. The governed-versus-local-only contrasts remain ecologically stable across the full ladder, while the hybrid-versus-top-down margin is the more contested part of the ordering.

the benchmark no longer asks only whether governance matters. It asks what central-only, local-only, and multi-scale governance each contribute under rising strategic pressure.

The resulting picture is more specific than a generic claim that "hybrid is best." Local-only coordination is not redundant with no governance; it improves some ecological outcomes relative to an unmanaged population. But it remains too weak to preserve the commons when adversarial pressure intensifies. Central control restores ecological stability, and hybrid governance then improves on central-only control enough to win every confirmatory decision cell. The main gain is ecological and behavioral rather than catastrophic. Garden failure is mostly saturated in the confirmatory rows, so the architecture result is carried by patch health, overharvest, and the incidence of restraint rather than by binary collapse alone.

The welfare analysis is equally important to the interpretation. Assessment of governance performance requires explicit attention to the cost of robust control. The incidence summaries show that the large welfare penalty in Harvest Commons is associated with moving away from local-only governance toward architectures that successfully restrain extraction. By contrast, the incremental welfare cost of moving from top-down-only to hybrid governance is moderate. This matters for the institutional reading of the result. It suggests that the benchmark is not simply identifying the most restrictive architecture. It is distinguishing between governance arrangements that differ in how they trade ecological protection against local burden.

The capability ladder further refines that interpretation. It shows that the strongest qualitative split in the benchmark is between architectures with central intervention and local-only governance. That ecological split survives the full non-LLM ladder. The more delicate question is whether hybrid governance preserves an advantage over top-down-only control as attacker strength rises. Here the answer is conditional rather than absolute: the hybrid margin often survives into the stronger rungs, but it is the part of the ordering most sensitive to attacker family, and its welfare threshold can be crossed earlier than its ecological edge disappears. This is still substantively informative. It identifies not only which architecture ranks first in the strongest confirmatory matrix, but also which comparison is robust and which comparison remains contestable as strategic pressure escalates.

At the same time, the scope of the claim remains deliberately narrow. The paper does not study all possible architectures, all possible attacker models, or all relevant notions of alignment. Harvest LLM remains outside the main evidence chain because it has not cleared the reliability gate. The environments remain

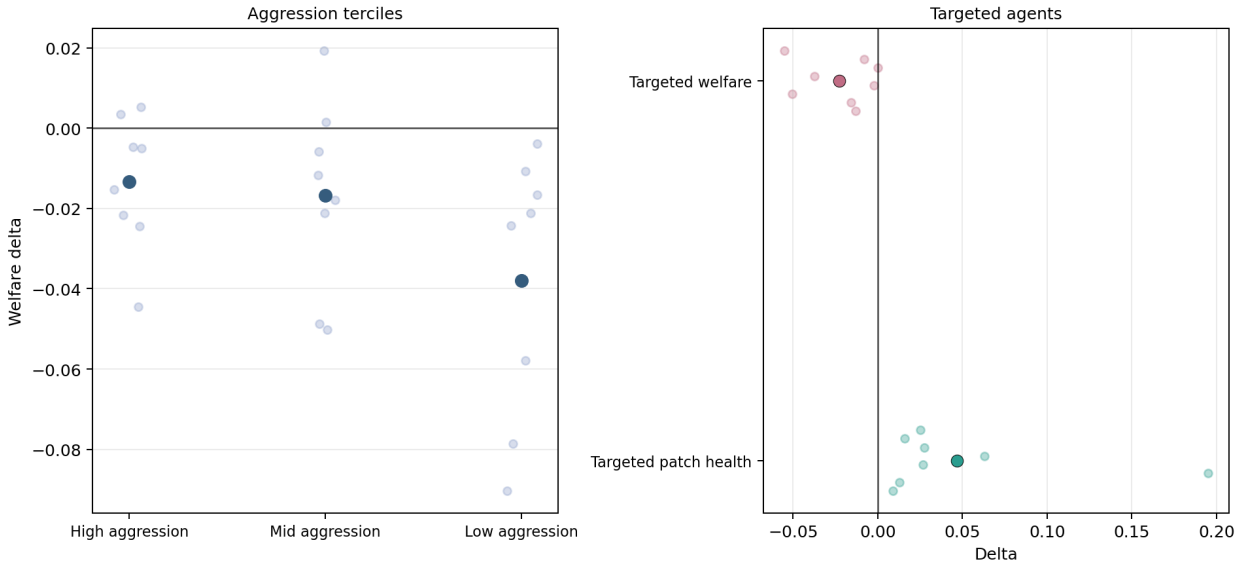

Figure 3: Welfare-incidence summary for Harvest Commons. The left panel shows cell-level welfare deltas by aggression tercile for `hybrid - top_down_only`; the larger markers indicate tercile means. The right panel shows the same contrast for targeted agents, reporting both welfare and local patch-health deltas. Relative to top-down-only control, hybrid governance imposes a modest incremental welfare cost while improving local patch health.

stylized, and the benchmark should be read as a tool for controlled comparison rather than as a direct model of deployment. Within those bounds, however, the follow-up supports a stronger empirical claim than the earlier draft: governance architecture in sequential commons can be evaluated as institutional robustness under capability escalation, and multi-scale governance can outperform central-only control without eliminating trade-offs.

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